1.

令 $\nu_n = \frac{1}{n}\sum_{i=1}^{n}\mu P^i$, 考虑

$$\nu_n P - \nu_n = \frac{1}{n}(\mu P^{n+1} - \mu P) \Rightarrow \|\nu_n P - \nu_n\|_1 < \frac{2}{n}.$$

因此 $\lim_{n\to\infty}\|\nu_n P - \nu_n\|_1 = 0$. 那么对任何收敛子列 $\{\nu_{n_k}\}$ 都有 $\lim_{k\to\infty}\nu_{n_k} = \pi$ 存在且唯一. ν 所在的有限维分布空间显然是紧的, 且上述子列存在且极限不依赖于子列选取. 最终 $\lim_{n\to\infty}\nu_n = \pi$.

2.

(1) 已证 $\pi P = \pi$ up to scaling 唯一, 因此 $\operatorname{rank}(\mathbf{1} - P) = n - 1$, $\lambda = 1$ 的几何重数为 1. 而如果存在 λ 的广义特征向量 v, 则存在 v 满足

$$(\mathbf{1} - P)v \neq 0, \quad (\mathbf{1} - P)^2 v = 0.$$

同上的 $P\mathbf{1}_{n\times 1} = \mathbf{1}_{n\times 1}$ up to scaling 唯一, 则 $u := (P - \mathbf{1})v = k\mathbf{1}_{n\times 1}$, 给出 $Pv = v + k\mathbf{1}_{n\times 1}$, 则 $\pi P v = \pi v = \pi v + k$, 但 $k \neq 0$, 矛盾. 所以这样的 v 不存在. λ 的代数重数也只能是 1.

(2) 注意到 $P^T|_{\Omega_i}$ 仍然是一个不可约 Markov 核, 有特征值 1, 它对应 P 的某个特征值 ω_k, 特征子空间维度为 1, 所以共 T 个 ω_k 两两不同, 给出所有 $\mathrm{e}^{2\pi\mathrm{i}\frac{k}{T}}$.

3.

显然 $f := \Pr[\exists i > 0,\ X_i = X_0] = p\beta + (1-p)\alpha$, 其中 $\beta = \Pr[\exists t,\ X_i = X_0 - 1]$, α 同理. 当 $p = \frac{1}{2}$, 显然 $f = 1$. 否则不妨 $p > \frac{1}{2}$, 由于 $\mathbb{E}[X_t - X_{t-1}] = 2p - 1 > 0$, $\lim_{t\to\infty}\mathbb{E}[X_t] = +\infty$, $\alpha = 1$. 设 $\beta_n = \Pr[\exists t, X_t = X_0 - n - 1]$ $(n \geq 0)$, $\beta_{-1} = 1$, $\beta_0 = \beta$, 而明显

$$\beta_n = p\beta_{n+1} + (1-p)\beta_{n-1} \Leftrightarrow p\beta_{n+1} = \beta_n + (p-1)\beta_{n-1},$$

特征方程 $pr^2 - r + (p-1) = 0$, 解得 $r = 1$ 或者 $r = \frac{1-p}{p}$, 结合 $\lim_{n\to\infty}\beta_n = 0$ 得到 $\beta_n = \beta\left(\frac{1-p}{p}\right)^n$, 同时

$$\beta_0 = 1 - p + p\beta_1 \Rightarrow \beta = \frac{1-p}{p}.$$

因而这时 $f = 2(1-p)$. $p < \frac{1}{2}$ 同理, 最终 $f = 2\min(p, 1-p)$.

4.

考虑提示中的

$$\begin{aligned}\mathbb{E}[\#\{k > 0 : X_k = X_0\}] &= \sum_{k=1}^{+\infty}\Pr[X_k = X_0] \\ &= \sum_{k=1}^{+\infty}\left(\Pr_{Y\sim\mathrm{Unif}(\{-1,0,1\})^k}\left[\sum_{i=1}^{k} X_i = 0\right]\right)^d.\end{aligned}$$

其中 $\mathbb{E}[X_i] = 0$, $\mathrm{Var}[X_i] = \frac{2}{3}$. 由中心极限定理, 取定任意 $\gamma' > 1$,

$$\exists K,\ \forall k \geq K,\ \Pr_X\left[\sum_{i=1}^{k} X_i = 0\right] \leq \frac{\gamma'}{\sqrt{2\pi k \cdot \frac{2}{3}}} =: \frac{\gamma}{\sqrt{k}}.$$

这时

$$\mathbb{E}[\#\{k > 0 : X_k = X_0\}] = \left(\sum_{i=1}^{K-1} + \sum_{i=K}^{+\infty}\right)\left(\Pr_{Y \sim \mathrm{Unif}(\{-1,0,1\})^k}\left[\sum_{i=1}^{k} X_i = 0\right]\right)^d.$$

对于 $d = 2$,

$$\mathbb{E}[\#\{k > 0 : X_k = X_0\}] \geq \sum_{i=K}^{+\infty} \frac{\gamma^2}{k} = +\infty.$$

而由 Markov 性质, 枚举第 j 次返回 X_0 的概率:

$$\mathbb{E}[\#\{k > 0 : X_k = 0\}] = \sum_{j=1}^{n} \Pr[\exists i > 0,\ X_i = X_0]^j.$$

所以只能 $\Pr[\exists i > 0,\ X_i = X_0] = 1$.

对于 $d \geq 3$ 时 $\mathbb{E}[\#\{k > 0 : X_k = X_0\}] < \infty$ 且 $\to 0$, 明所欲证.

5.

只需要证明 $d(2\tau(\varepsilon)) \leq 2\varepsilon^2$. 对任意分布 x, 设 $\mu = xP^{\tau(\varepsilon)}$, 则 $\|\mu - \pi\|_1 \leq 2\varepsilon$. 设

$$\begin{cases} \mu^+ = \max\{0, \mu - \pi\}, \\ \mu^- = \max\{0, \pi - \mu\}. \end{cases}$$

有 $\|\mu^+\|_1 \leq \varepsilon$, $\|\mu^-\|_1 \leq \varepsilon$. 则

$$\begin{aligned} \|\mu P^{\tau(\varepsilon)} - \pi\|_1 &= \|(\mu^+ - \mu^-)P^{\tau(\varepsilon)}\|_1 \\ &\leq \varepsilon\|\widehat{\mu^+}P^{\tau(\varepsilon)} - \pi\|_1 + \varepsilon\|\pi - \widehat{\mu^-}P^{\tau(\varepsilon)}\|_1 \\ &\leq 2\varepsilon \cdot 2\varepsilon = 4\varepsilon^2. \end{aligned}$$

即 $d(2\tau(\varepsilon)) \leq \frac{1}{2} \cdot 4\varepsilon^2 = 2\varepsilon^2$.

6.

考虑 $d(\varepsilon) \leq \overline{d}(\varepsilon) = \sup_{\delta_X, \delta_Y}\{t : \Delta_{\mathrm{TV}}(\delta_X P^t, \delta_Y P^t) \leq \varepsilon\}$, 尝试用 coupling 来限制后者. 对 $(X, Y) \in (V \to C)^2$, 考虑 Markov 核 P_{XY}, 其过程如下:

- 均匀随机选取 $u \in V$:
  - 令 $A = C \setminus \{X(v) : (u, v) \in E\}$, $B = C \setminus \{Y(v) : (u, v) \in E\}$;
  - 以 $\frac{|A \cap B|}{\max(|A|, |B|)}$ 的概率:
    - 均匀随机取 $c \in A \cap B$, 令 $X(u) \leftarrow c$, $Y(u) \leftarrow c$;
  - 以剩余概率:
    - 按适当分布取 $(c_X, c_Y) \in A \times B$, 令 $X(u) \leftarrow c_X$, $Y(u) \leftarrow c_Y$, 使得边缘分布与原 Markov 核相同.

若 $\Delta_{\mathrm{Hamming}}(X, Y) = 1$, 设 $X(w) \neq Y(w)$, 讨论 $u = w$, $(u, w) \in E$ 和其他的 u 共三种情况, 可知

$$
\begin{aligned}
\mathbb{E}[\Delta_{\text{Hamming}}(X', Y')] &\leq \frac{n-1}{n} + \frac{\Delta}{n}\left(1 - \frac{C-\Delta-1}{C-\Delta}\right) \\
&= 1 - \frac{1}{n} + \frac{1}{4n} \\
&= 1 - \frac{3}{4n}.
\end{aligned}
$$

一般地, 设 $\Delta_{\text{Hamming}}(X, Y) = k$, 任取 $\{X = Z_0, \cdots, Z_k = Y\}$ 满足 $\Delta_{\text{Hamming}}(Z_i, Z_{i+1}) = 1$, 则

$$
\begin{aligned}
\mathbb{E}[\Delta_{\text{Hamming}}(X', Y')] &\leq \mathbb{E}\left[\sum_{i=1}^{k} \Delta_{\text{Hamming}}(Z'_{i-1}, Z'_i)\right] \\
&= \sum_{i=1}^{k} \mathbb{E}[\Delta_{\text{Hamming}}(Z'_{i-1}, Z'_i)] \\
&\leq \sum_{i=1}^{k} \left(1 - \frac{3}{4n}\right) \\
&= \left(1 - \frac{3}{4n}\right) \Delta_{\text{Hamming}}(X, Y).
\end{aligned}
$$

因此

$$
\mathbb{E}_{(X_0,Y_0)}[\Delta_{\text{Hamming}}(X_t, Y_t)] = \left(1 - \frac{3}{4n}\right)^t \Delta_{\text{Hamming}}(X_0, Y_0) \leq n\left(1 - \frac{3}{4n}\right)^t < n\mathrm{e}^{-\frac{3t}{4n}}.
$$

取 $T = \left\lceil \frac{4}{3} n \ln \frac{n}{\varepsilon} \right\rceil$, 就有

$$
\begin{aligned}
d(\varepsilon) &\leq \sup_{\delta_{X_0}, \delta_{Y_0}} \left\{ t : \Pr_{(X_0,Y_0)}[X_t \neq Y_t] \leq \varepsilon \right\} \\
&\leq \sup_{\delta_{X_0}, \delta_{Y_0}} \left\{ t : \Pr_{(X_0,Y_0)}[\Delta_{\text{Hamming}}(X_t, Y_t) \geq 1] \leq \varepsilon \right\} \\
&\leq \sup_{\delta_{X_0}, \delta_{Y_0}} \left\{ t : \mathbb{E}_{(X_0,Y_0)}[\Delta_{\text{Hamming}}(X_t, Y_t)] \leq \varepsilon \right\} \\
&\leq \sup_{\delta_{X_0}, \delta_{Y_0}} \left\{ t : n\mathrm{e}^{-\frac{3t}{4n}} \leq \varepsilon \right\} \\
&\leq T.
\end{aligned}
$$